Unive Mentorship Program Summer 2026

Real-Time Transaction Risk & Velocity Enforcement Engine for Point-of-Sale Terminals

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Project Title

Real-Time Transaction Risk & Velocity Enforcement Engine for Point-of-Sale Terminals

Project Objective

Design, verify, and physically implement (RTL → GDSII) a fully synthesizable digital IP that performs real-time, rule-based fraud risk screening on a stream of point-of-sale transaction records, computing rolling-window velocity and cumulative-spend checks entirely in hardware, using the OpenROAD flow on IHP SG13G2 130nm.

Problem Statement

Point-of-sale and ATM fraud detection is typically handled by backend servers, introducing network latency and dependence on connectivity. A dedicated local hardware pre-check block can immediately flag suspicious patterns — rapid repeated transactions, spend exceeding a configured limit, abnormal transaction amounts — before or alongside backend authorization, adding a deterministic, low-power layer of protection suited to embedded or low-cost payment terminals.

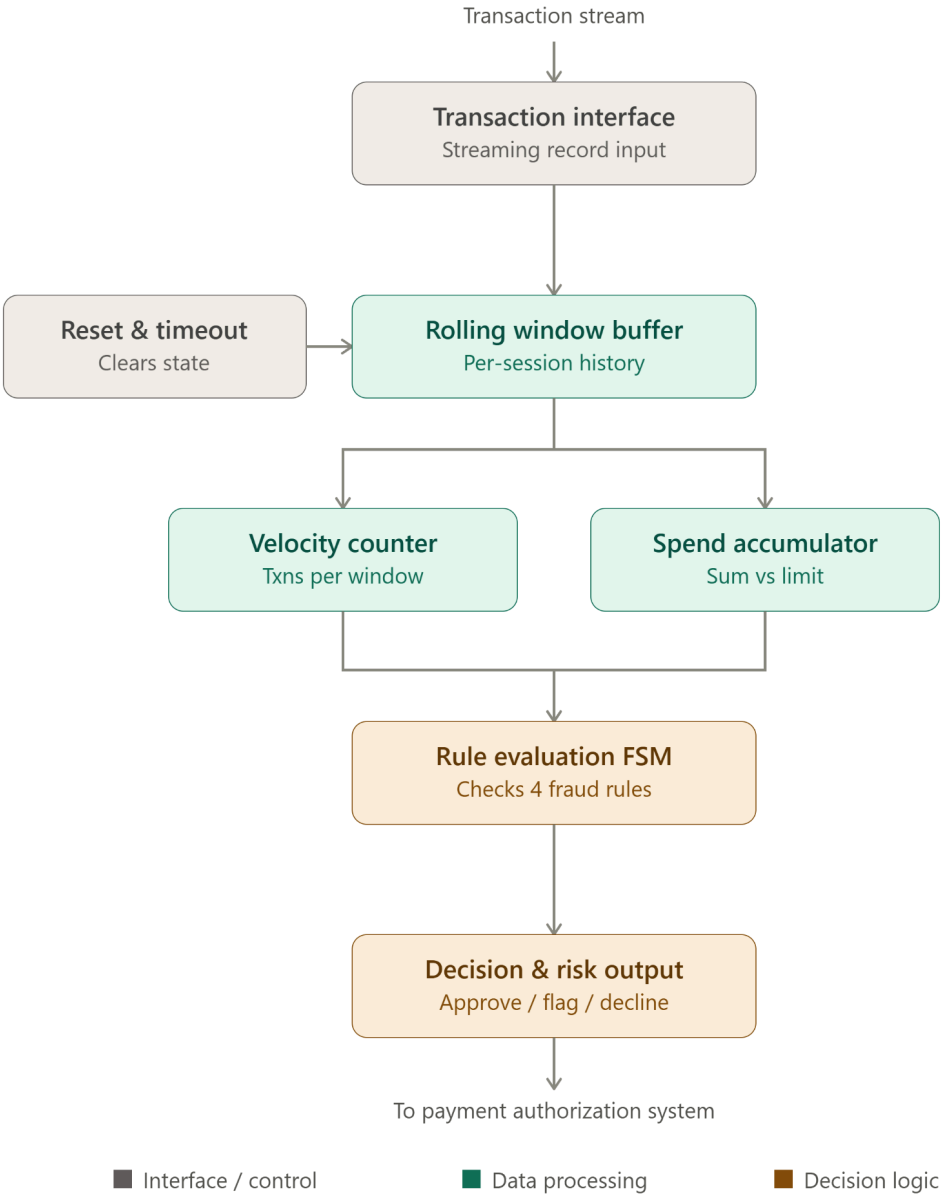
Application Domain

FinTech & Digital Payments

Project Overview

The IP receives a stream of transaction records (amount, category code, time-delta since the previous transaction) one per cycle via a valid handshake. A small flip-flop-based rolling window tracks the last few transactions per active session. A velocity counter tracks transaction count within a configurable time window; a spend accumulator sums amounts within that window against a programmable limit. A rule-evaluation FSM checks for over-limit spend, high velocity, rapid-repeat transactions, and amount anomalies, producing a risk score and a three-way decision (approve / flag-for-review / decline).

Block Diagram



Features

1. Streaming transaction interface, one record per cycle
2. Rolling-window velocity counter and cumulative-spend tracker (flip-flop based, no SRAM)
3.Programmable time window, transaction-count limit, spend limit, and anomaly threshold
4.Fixed rule-based risk engine: over-limit, high-velocity, rapid-repeat, amount-anomaly checks
5.Three-way decision output (approve / flag / decline) plus numeric risk score
6.Session reset/timeout handling
7.Fully synchronous, flat RTL, no memory macros
8.Self-checking testbench: normal spending, velocity abuse, over-limit, and edge cases

Design Specifications

Parameter	Value
Technology	IHP SG13G2 (130nm), open PDK
Language	Verilog HDL (synthesizable)
Flow	OpenROAD-flow-scripts (ORFS)
Target Clock	20.0 ns period (50 MHz)
Transaction Input	16-bit amount, 8-bit category code, 12-bit time-delta, valid strobe
Control Inputs	reset, window size, count limit, spend limit, anomaly threshold
Outputs	decision[1:0], risk_score[7:0], valid_out
Memory Macros	None — registers and combinational logic only